

Name:

Maths Class:

Year 12
Mathematics Extension 2

HSC Course
Trial Examination

August, 2022

Time allowed: 3 hours + 10 minutes reading time

General Instructions:

- Marks for each question are indicated on the question.
- NESA approved calculators may be used
- In Questions 11-16, show all relevant mathematical reasoning and/or calculations
- Full marks may not be awarded for careless work or illegible writing
- ***Begin each question on a new page***
- Write using black pen
- All answers are to be in the writing booklet provided
- A reference sheet is provided.

Total marks – 70

Section I - 10 Marks

- Attempt Question 1-10 on the sheet provided
- Allow about 15 minutes for this section

Section II – 90 Marks

- Attempt Questions 11-14
- Allow about 2 hours and 45 minutes for this section

Section I

10 Marks

Attempt Questions 1-10

Allow about 15 minutes for this section

Use the multiple choice answer sheet for Questions 1-10.

1. Consider the statement:

$$\forall x: |2x + 5| \leq 9 \Rightarrow |x| \leq 4$$

Which of the following is a counterexample to this statement?

- (A) $x = 0$
- (B) $x = -6$
- (C) $x = -10$
- (D) $x = 10$

2. What is the magnitude of the vector $(\cos \theta)\underline{\hat{i}} + (\sin \theta)\underline{\hat{j}} + (\tan \theta)\underline{\hat{k}}$?

- (A) 1
- (B) $\operatorname{cosec} \theta$
- (C) $\cot \theta$
- (D) $\sec \theta$

3. Where $\lambda \in \mathbb{R}$, which of the following is a vector equation of the line through $(5,3)$, parallel to $2\underline{\hat{i}} + 4\underline{\hat{j}}$?

- (A) $2\underline{\hat{i}} + 4\underline{\hat{j}} + \lambda(5\underline{\hat{i}} + 3\underline{\hat{j}})$
- (B) $2\underline{\hat{i}} + 4\underline{\hat{j}} + \lambda(3\underline{\hat{i}} - 5\underline{\hat{j}})$
- (C) $5\underline{\hat{i}} + 3\underline{\hat{j}} + \lambda(\underline{\hat{i}} + 2\underline{\hat{j}})$
- (D) $5\underline{\hat{i}} + 3\underline{\hat{j}} + \lambda(2\underline{\hat{i}} - \underline{\hat{j}})$

4. Given $z = 3\left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right)$, which of the following is $(\bar{z})^{-1}$ in polar form?

(A) $\frac{1}{3}\left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right)$

(B) $-\frac{1}{3}\left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right)$

(C) $\frac{1}{3}\left(\cos\left(-\frac{\pi}{6}\right) + i\sin\left(-\frac{\pi}{6}\right)\right)$

(D) $-\frac{1}{3}\left(\cos\left(-\frac{\pi}{6}\right) + i\sin\left(-\frac{\pi}{6}\right)\right)$

5. Which of the following is equivalent to i^i ?

(A) $e^{-\frac{\pi}{2}}$

(B) $e^{\frac{\pi}{2}}$

(C) i

(D) $-i$

6. State the values of A , B , and C if

$$\frac{4x + 10}{(2 - x)(x^2 + 2)} = \frac{A}{2 - x} + \frac{Bx + C}{x^2 + 2}$$

(A) $A = 3, B = -1, C = 5,$

(B) $A = 3, B = 3, C = 2$

(C) $A = 5, B = -1, C = 0$

(D) $A = 5, B = -6, C = 5$

7. Consider the statement: “If I have power, then I have responsibility.”

Which of the following is NOT an equivalent statement?

(A) If I do not have responsibility, then I do not have power.

(B) If I have responsibility, then I have power.

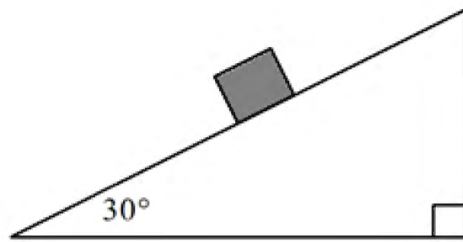
(C) Either I do not have power, or I have responsibility.

(D) I cannot have both power and no responsibility.

8. The velocity of a particle varies with respect to its displacement, such that $v = 9 - 6x - x^2$, what is the acceleration when $x = 0$?

- (A) 6
- (B) -6
- (C) 54
- (D) -54

9. An object is placed on a smooth ramp inclined at an angle of 30° to the horizontal.



If the acceleration due to gravity is 10 ms^{-2} , at what velocity is the object travelling down the ramp after t seconds?

- (A) $5t \text{ ms}^{-1}$
- (B) 5 ms^{-1}
- (C) $5\sqrt{3}t \text{ ms}^{-1}$
- (D) $5\sqrt{3} \text{ ms}^{-1}$

10. Which of the following integrals is greater than zero?

(A)

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} e^{x^2} \cos x \, dx$$

(B)

$$\int_{-\pi}^{\pi} x^3 \cos x \, dx$$

(C)

$$\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} (\sin^2 x - \cos^2 x) \, dx$$

(D)

$$\int_{-1}^1 \sin^{-1}(x^3) \, dx$$

Section II

Total marks – 90

Attempt Question 11-16

Allow about 2 hour and 45 minutes for this section

Begin each question on a NEW page

In Questions 11-14, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Begin a NEW page.

a) If $z = 5 + i$ and $w = 4 - 2i$, simplify in $x + iy$ form

i) $z + iw$ 1

ii) $z\bar{w}$ 1

iii) $\frac{w}{z}$ 2

b) For the complex number $z = -3 - 3i$

i) Express z in the form $r(\cos \theta + i \sin \theta)$. 2

ii) Hence or otherwise, find the smallest positive integer n , such that $z^n \in \mathbb{R}$. 2

c) Prove that $x^2 + x + 1$ is odd for all positive integer values of x . 2

d) If $\underline{a} = \underline{i} + 2\underline{j} + 3\underline{k}$ and $\underline{b} = 4\underline{i} + 5\underline{j} + 6\underline{k}$, evaluate $(\underline{a} - 2\underline{b}) \cdot \underline{a}$. 1

e) Find the vector $\underline{v}$ parallel to $\underline{u} = 3\underline{i} - 2\underline{j} + 4\underline{k}$ that has a magnitude of 3. 2

f) Find: 2

$$\int \frac{2x + 3}{x + 1} dx$$

End of Question 11

Question 12 (15 marks) Begin a NEW page.

a) Where $z = x + iy$, sketch the graph defined by $\text{Im}(z^2) = 4$. 2

b) Find the following integrals

i) 2

$$\int \frac{1}{\sqrt{5 + 4x - x^2}} dx$$

ii) 3

$$\int \sin^4 x \cos^3 x \, dx$$

c) Evaluate 2

$$\int_1^{\sqrt{5}} x^3 \sqrt{x^2 - 1} \, dx$$

d) A particle is moving in a straight line in simple harmonic motion. The amplitude of the motion is 4 cm, and the period of the motion is 8 seconds. The particle begins at the maximum displacement from the centre of motion.

i) Find when the particle is first 2 cm to the right of the centre of motion. 2

ii) Calculate the maximum velocity of the particle. 2

iii) Find the maximum acceleration, and the first time at which this occurs. 2

End of Question 12

Question 13 (15 marks) Begin a NEW page.

a) It is known that $P(x) = 2x^3 - 19x^2 + 112x + d$, where $d \in \mathbb{R}$, has a zero of $5 + 6i$.

i) What are the two other zeroes of $P(x)$? 2

ii) Find the value of d . 2

b) If ω is a non-real cube root of unity, prove that:

i) $1 + \omega + \omega^2 = 0$ 1

ii) $(1 + \omega^2)(1 + \omega - \omega^2)^7 = 128$ 2

c) For integers a, b, c , with $b, c \leq 9$, the number x can be written as $100a + 10b + c$. 3
For example, the number 56789 can be written as $100 \times 567 + 10 \times 8 + 9$.

Use this to prove that x is divisible by 4 **if and only if** the last two digits are divisible by 4.

d) Let $\vec{a} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$ and $\vec{b} = \begin{bmatrix} 2 \\ -4 \\ 4 \end{bmatrix}$. If θ is the acute angle between these vectors, find 3
the exact value of $\sin 2\theta$.

e) Find the point of intersection of the vector lines 2

$$\vec{r} = \begin{bmatrix} 0 \\ 3 \\ -5 \end{bmatrix} + \lambda \begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix} \text{ and } \vec{q} = \begin{bmatrix} -1 \\ 2 \\ 3 \end{bmatrix} + \mu \begin{bmatrix} -1 \\ 2 \\ -1 \end{bmatrix}$$

End of Question 13

Question 14 (15 marks) Begin a NEW page.

- a) A particle moves with equation of motion $x = \sin 3t + \cos 3t + 2$ metres. 2
Prove that the particle is moving in simple harmonic motion and find the centre and period of its motion.

- b) For integers a and b , with $b \neq 0$, consider the statement:

“If x is irrational, then $a + bx$ is irrational.”

- i) State the contrapositive of this statement. 1
- ii) Hence, or otherwise, show that the statement is true. 2
- iii) Show that $\sqrt{3}$ is irrational, and hence deduce that $5 + 7\sqrt{3}$ is irrational. 2
- c) Determine whether the origin $(0, 0, 0)$ lies inside, outside, or on the sphere given by the equation $x^2 - 2x + y^2 + z^2 + 4z + 4 = 0$. 2
- d) The line L_1 is given by $\tilde{r} = \begin{bmatrix} 1 \\ -2 \\ 4 \end{bmatrix} + \lambda \begin{bmatrix} 4 \\ 5 \\ -6 \end{bmatrix}$ and the line L_2 passes the points 2
 $(3, 8, -2)$ and $(a, -2, 10)$. Find the value of a if L_1 is parallel to L_2 .

- e) Where a is a positive constant

- i) Show that 2

$$\int_0^a f(x) dx = \int_0^a f(a-x) dx$$

- ii) Hence, or otherwise, evaluate 2

$$\int_0^1 x(1-x)^{50} dx$$

End of Question 14

Question 15 (15 marks) Begin a NEW page.

- a) The acceleration of a particle is given by $\ddot{x} = -\frac{1}{2}e^{-4x}$.

Initially, the particle is at the origin with a velocity of $\dot{x} = \frac{1}{2}ms^{-1}$.

- i) Express velocity as a function of displacement. 2
- ii) Find the displacement as a function of time. 2
- iii) Describe the limiting behaviour of displacement and velocity (as $t \rightarrow \infty$). 1

- b) For the integral

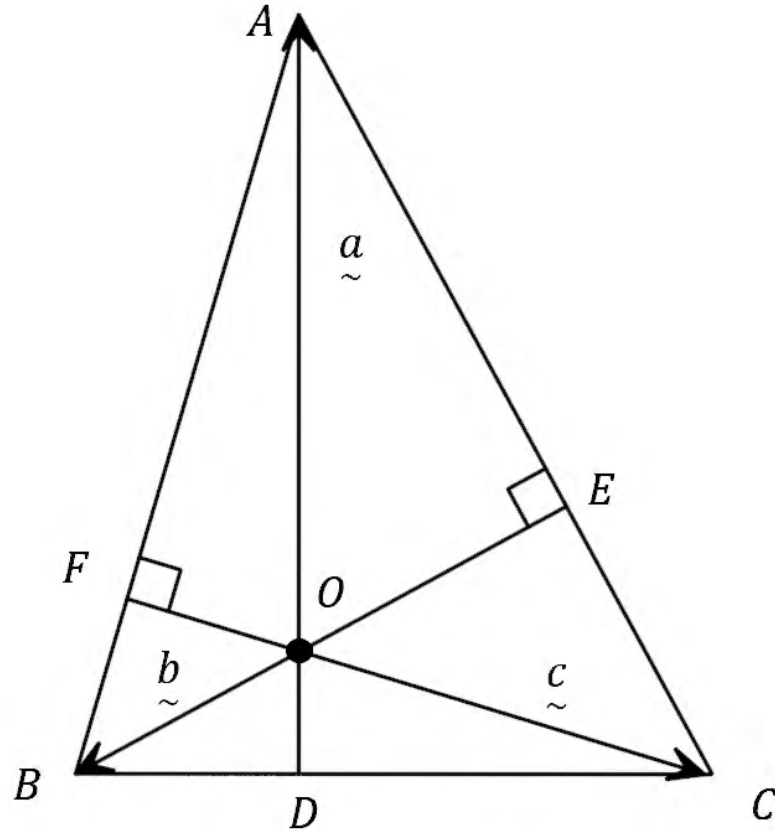
$$I_n = \int_0^{\frac{\pi}{2}} x^n \cos x \, dx$$

- i) Show that $I_n = \left(\frac{\pi}{2}\right)^n - n(n-1)I_{n-2}$ 3
- ii) Hence, find the value of I_4 in terms of π . 2

Question 15 continues on Page 10

- c) In the diagram below, $\vec{a} = \overrightarrow{OA}$, $\vec{b} = \overrightarrow{OB}$, and $\vec{c} = \overrightarrow{OC}$.

$\triangle ABC$ is an acute angled triangle. O is the point of intersection of altitudes CF and BE (i.e. $BE \perp AC$ and $CF \perp AB$). $\overrightarrow{AO}$ is produced from O to point D on BC .



- i) Show that $\vec{b} \cdot (\vec{c} - \vec{a}) = 0$ and $\vec{c} \cdot (\vec{b} - \vec{a}) = 0$. 2
- ii) Hence, show that $AD \perp BC$. 2
- iii) What geometrical property of the triangle has been proven? 1

End of Question 15

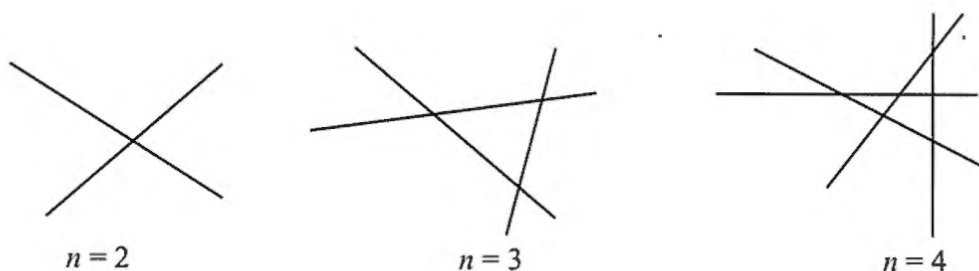
Question 16 (15 marks) Begin a NEW page.

a)

i) Show that $(1 + i \tan \theta)^n + (1 - i \tan \theta)^n = \frac{2 \cos n\theta}{\cos^n \theta}$ 2
Where $\cos \theta \neq 0$ and n is a positive integer.

ii) Hence show that if z is a purely imaginary number, then the roots of $(1 + z)^4 + (1 - z)^4 = 0$ are $z = \pm i \tan \frac{\pi}{8}, \pm i \tan \frac{3\pi}{8}$. 2

b) Consider n lines drawn on a plane, as shown in the series of diagrams below. Let X_n be the maximum number of possible points of intersection of these lines.



i) Find the values of a , b , and c in the table below 1

n	2	3	4	5	6
X_n	1	3	a	b	c

ii) Prove by mathematical induction that for $n \geq 2$, X_n is given by 3

$$X_n = \frac{n(n-1)}{2}$$

c) Prove the following, where x, a, b, c are real positive numbers.

i) $x + \frac{1}{x} \geq 2$ 2

ii) Hence, show $(a + b + c) \left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \right) \geq 9$ 2

iii) Hence, show $\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} \geq \frac{3}{2}$ 3

End of Examination

1. (B) $|2x - 6 + 5| = 7 \checkmark$

$| -6 | > 4 \quad \times$

2. (D) $\sqrt{\cos^2 \theta + \sin^2 \theta + \tan^2 \theta} = \sqrt{1 + \tan^2 \theta}$
 $= \sec \theta$

3. (C) as $\underline{i} + 2\underline{j}$ is parallel to $2\underline{i} + 4\underline{j}$

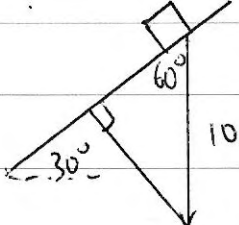
4. (A) $\frac{1}{3(\cos \frac{\pi}{6} - i \sin \frac{\pi}{6})} = \frac{1}{3} \frac{\cos \frac{\pi}{6} + i \sin \frac{\pi}{6}}{(\cos^2 \frac{\pi}{6} + \sin^2 \frac{\pi}{6})} =$

5. (A) $i^i = (e^{\frac{\pi}{2}i})^i = e^{-\frac{\pi}{2}}$

6. (B) $4x + 10 = A(x^2 + 2) + (Bx + C)(2 - x)$
At $x = 2$ At $x = 0$
 $18 = A \cdot 6$ $10 = 3 \cdot 2 + C \cdot 2$
 $A = 3$ $C = 2$

7. (B) Converse is not equivalent.

8. (D) $a = v \times \frac{dv}{dx} = (-6 - 2x)(9 - 6x - x^2)$
 $\therefore$ At $x = 0$ $a = -6 \times 9 = -54$

9. (A)  $a = 10 \cos 60^\circ$
 $= 5$
 $\therefore v = 5t$

10. (A) $e^{x^2} \cos x \geq 0$ and even, in $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$.

1 11.

2 a) $z = 5 + i$, $w = 4 - 2i$

3

4 i) $z + iw = 5 + i + i(4 - 2i)$
 5 $= 5 + i + 4i + 2$
 6 $= 7 + 5i$

7

8 ii) $z\bar{w} = (5 + i)(4 + 2i)$
 9 $= 20 + 10i + 4i - 2$
 10 $= 18 + 14i$

11

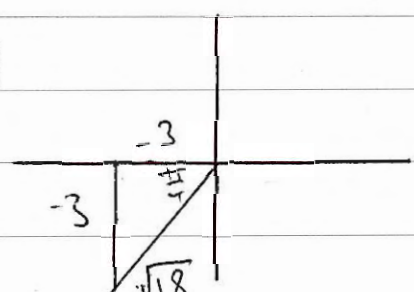
12 iii) $\frac{w}{z} = \frac{4 - 2i}{5 + i} \times \frac{5 - i}{5 - i}$
 13 $= \frac{20 - 10i - 4i - 2}{25 + 1}$
 14 $= \frac{18}{26} - \frac{14}{26}i = \frac{9}{13} - \frac{7}{13}i$

17

18 b) $z = -3 - 3i$

19

20 i) $\therefore z = 3\sqrt{2} \left(\cos\left(-\frac{3\pi}{4}\right) + i\sin\left(-\frac{3\pi}{4}\right) \right)$



24

25 ii) $\arg(z) = -\frac{3\pi}{4}$
 26 As $\sin(k\pi) = 0$, $k \in \mathbb{Z}$, $n = 4$.

$$c) x^2 + x + 1$$

$$= x(x+1) + 1$$

If x is odd, $x+1$ is

even, so

$x(x+1) + 1$ is odd

If x is even

$x(x+1) + 1$ is odd.

$$d) (\underline{a} - 2\underline{b}) \cdot \underline{a}$$

$$= (\underline{i} + 2\underline{j} + 3\underline{k} - 2(\underline{4i} + \underline{5j} + \underline{6k})) \cdot (\underline{i} + 2\underline{j} + 3\underline{k})$$

$$= (-\underline{7i} - \underline{8j} + \underline{9k}) \cdot (\underline{i} + 2\underline{j} + 3\underline{k})$$

$$= -7 - 16 - 27$$

$$= -50$$

$$e) |\underline{u}| = \sqrt{29}$$

$$\therefore \underline{u} = \frac{3}{\sqrt{29}} (\underline{3i} - \underline{2j} + \underline{4k})$$

$$f) \int \frac{2x+3}{x+1} dx = \int \frac{2x+2+1}{x+1} dx$$

$$= \int 2 + \frac{1}{x+1} dx$$

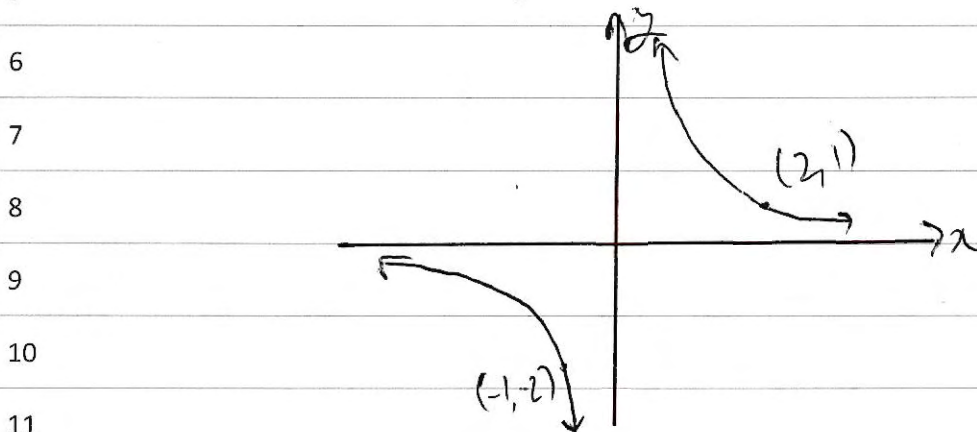
$$= 2x + \ln|x+1| + c$$

1 12. a) $\text{Im}(x+iy)^2 = 4$

2 $\text{Im}(x^2 - y^2 + 2xyi) = 4$

3 $2xy = 4$

4 $xy = 2$



12

13 b) i) $\int \frac{1}{\sqrt{5+4x-x^2}} dx$

14

15

16 $= \int \frac{1}{\sqrt{5-(x^2-4x+4)}} dx$

17

18 $= \int \frac{1}{\sqrt{9-(x-2)^2}} dx$

19

20 $= \sin^{-1}\left(\frac{x-2}{3}\right) + C$

21

22

23

24

25

26

1

2

$$ii) \int \sin^4 x \cos^3 x \, dx$$

3

4

$$= \int \sin^4 x \cos^3 x \cos x \, dx$$

$$u = \sin x$$

$$du = \cos x \, dx$$

5

6

$$= \int u^4 (1-u^2) \, du$$

7

8

$$= \int u^4 - u^6 \, du$$

9

10

$$= \frac{1}{5} u^5 - \frac{1}{7} u^7 + C$$

11

12

$$= \frac{1}{5} \sin^5 x - \frac{1}{7} \sin^7 x + C.$$

13

14

$$c) \int_1^{\sqrt{5}} x^3 \sqrt{x^2-1} \, dx$$

$$u = x^2 - 1 \quad x^2 = u + 1$$

15

$$x = 1 \rightarrow u = 0$$

16

$$= \frac{1}{2} \int_1^{\sqrt{5}} 2x x^2 \sqrt{x^2-1} \, dx$$

$$x = \sqrt{5} \rightarrow u = 4$$

17

$$du = 2x \, dx$$

18

$$= \frac{1}{2} \int_0^4 (u+1) \sqrt{u} \, du$$

19

20

$$= \frac{1}{2} \int_0^4 (u+1) u^{\frac{1}{2}} \, du$$

21

22

$$= \frac{1}{2} \int_0^4 u^{\frac{3}{2}} + u^{\frac{1}{2}} \, du$$

23

24

$$= \frac{1}{2} \left[\frac{2}{5} u^{\frac{5}{2}} + \frac{2}{3} u^{\frac{3}{2}} \right]_0^4$$

25

26

$$= \frac{1}{2} \left(\frac{2}{5} \times 2^5 + \frac{2}{3} \times 2^3 - 0 \right)$$

$$= \frac{136}{15}$$

$$d) \quad a = 4 \quad \frac{2\pi}{n} = 8 \quad \text{At } t=0, \quad x=4.$$

$$n = \frac{\pi}{4}$$

$$\therefore \text{ Use } x = 4 \cos\left(\frac{\pi}{4}t\right)$$

$$i) \quad \text{When } x=2.$$

$$2 = 4 \cos\left(\frac{\pi}{4}t\right)$$

$$\cos\left(\frac{\pi}{4}t\right) = \frac{1}{2}$$

$$\frac{\pi}{4}t = \frac{\pi}{3}$$

$$t = \frac{4}{3}$$

$$ii) \quad \dot{x} = -4 \times \frac{\pi}{4} \sin\left(\frac{\pi}{4}t\right)$$

$$= -\pi \sin\left(\frac{\pi}{4}t\right)$$

$$\therefore \text{ Max } \dot{x} = -\pi \times -1$$

$$= \pi \text{ cm/s.}$$

$$iii) \quad \ddot{x} = -\frac{\pi^2}{4} \cos\left(\frac{\pi}{4}t\right)$$

$$\therefore \text{ Max acceleration is } \frac{\pi^2}{4}, \text{ occurs at}$$

$$\cos\left(\frac{\pi}{4}t\right) = -1$$

$$\frac{\pi}{4}t = \pi$$

$$t = 4$$

$$13. a) i) P(x) = 2x^3 - 19x^2 + 112x + d.$$

$$2 \quad \text{Zeros } \alpha = 5 + 6i, \beta = 5 - 6i, \gamma$$

$$4 \quad \alpha + \beta + \gamma = \frac{19}{2}$$

$$5 \quad 5 + 6i + 5 - 6i + \gamma = \frac{19}{2}$$

$$6 \quad \gamma + 10 = \frac{19}{2}$$

$$7 \quad \gamma = -\frac{1}{2}.$$

$$9 \quad \text{iii) } \alpha\beta\gamma = \frac{-d}{2}$$

$$10 \quad + \frac{1}{2} (5 - 6i)(5 + 6i) = \frac{-d}{2}$$

$$11 \quad d = 25 + 36$$

$$12 \quad = 61$$

$$14 \quad b) i) w \text{ solves } z^3 - 1 = 0$$

$$15 \quad (z - 1)(z^2 + z + 1) = 0$$

$$16 \quad w - \text{non real, so it solves } z^2 + z + 1 = 0$$

$$17 \quad \therefore w^2 + w + 1 = 0.$$

$$19 \quad \text{ii) From i)}$$

$$20 \quad 1 + w^2 = -w \quad (1)$$

$$20 \quad 1 + w - w^2 = -2w^2 \quad (2)$$

$$22 \quad (1) \times (2)^7: (1 + w^2)(1 + w - w^2)^7 = -w \times (-2w^2)^7$$

$$23 \quad = -w \times -128w^{14}$$

$$24 \quad = 128w^{5 \times 3}$$

$$25 \quad = 128$$

26

1

2 c) $\Rightarrow$: If x is divisible by 4, $x = 4M$, $M \in \mathbb{Z}$.

3

$$4 \quad \therefore 100a + 10b + c = 4M$$

$$5 \quad 10b + c = 4M - 100a$$

$$6 \quad = 4(M - 25a)$$

$$7 \quad = 4L, \quad L \in \mathbb{Z}.$$

8 $\therefore 10b + c$ is divisible by 4, with $10b + c$ as
9 the last two digits.

10

11 $\Leftarrow$: If the last two digits are divisible
12 by 4, $10b + c = 4k$, $k \in \mathbb{Z}$.

13

$$14 \quad \therefore x = 100a + 10b + c$$

$$15 \quad = 100a + 4k$$

$$16 \quad = 4(25a + k)$$

$$17 \quad = 4m, \quad m \in \mathbb{Z}.$$

18

19 $\therefore x$ is divisible by 4.

20

21 $\therefore x$ is divisible by 4 $\Leftrightarrow$ The last two
22 digits are divisible
23 by 4.

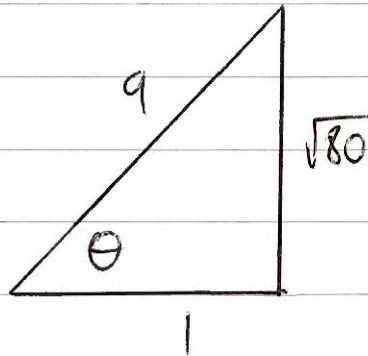
24

25

26

$$\begin{aligned} d) \quad \underbrace{a \cdot b}_{=2} &= 2 - 8 + 8 & \underbrace{|a|}_{=3} &= \sqrt{9} & \underbrace{|b|}_{=6} &= \sqrt{36} \\ & & & & & \end{aligned}$$

$$\begin{aligned} \therefore 3 \times 6 \times \cos \theta &= 2 \\ \cos \theta &= \frac{1}{9} \end{aligned}$$



$$\begin{aligned} \therefore \sin 2\theta &= 2 \sin \theta \cos \theta \\ &= 2 \times \frac{\sqrt{80}}{9} \times \frac{1}{9} \\ &= \frac{8\sqrt{5}}{81} \end{aligned}$$

1

$$2 \quad e) \text{ If } \begin{bmatrix} 0 \\ 3 \\ -5 \end{bmatrix} + \lambda \begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \\ 3 \end{bmatrix} + \mu \begin{bmatrix} -1 \\ 2 \\ -1 \end{bmatrix}$$

4

5

6

$$\begin{bmatrix} -\lambda \\ \lambda \\ 2\lambda \end{bmatrix} = \begin{bmatrix} -\mu - 1 \\ 2\mu - 1 \\ -\mu + 8 \end{bmatrix}$$

7

8

$$\therefore -\lambda = -\mu - 1 \quad \text{--- (1)}$$

9

$$\lambda = 2\mu - 1 \quad \text{--- (2)}$$

10

$$2\lambda = -\mu + 8 \quad \text{--- (3)}$$

11

12

$$(1) + (2): 0 = \mu - 2 \quad \text{--- (3)}$$

13

14

$$(3) \rightarrow (2) \quad 2 \times 2 - 1 = \lambda$$

15

$$\lambda = 3.$$

16

17

Test in (3)

18

$$LHS = 6$$

$$RHS = -2 + 8$$

19

$$= 6 \quad \checkmark$$

20

21

$$\therefore \text{Point at } \lambda = 3: \begin{bmatrix} 0 \\ 3 \\ -5 \end{bmatrix} + 3 \begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix}$$

22

23

$$= \begin{bmatrix} -3 \\ 6 \\ 1 \end{bmatrix}$$

24

25

26

$$\therefore \text{Point } (-3, 6, 1)$$

1 14

2 a) $x = \sin 3t + \cos 3t + 2$

3 $\dot{x} = 3\cos 3t - 3\sin 3t$

4 $\ddot{x} = -9\sin 3t - 9\cos 3t$

5 $= -9(\sin 3t + \cos 3t + 2 - 2)$

6 $= -9(x - 2)$

7

8 $\therefore$ SHM with centre of motion $x=2$
 9 and period $\frac{2\pi}{3}$.

10

11 b) i) If $a+bx$ is rational, x is rational.

12

13 ii) If $a+bx$ is rational:

14

15 $a+bx = \frac{p}{q} \quad p, q \in \mathbb{Z}, q \neq 0.$

16

17 $bx = \frac{p}{q} - a$

18

19 $x = \frac{p - qa}{qb} \quad \begin{matrix} p - qa \in \mathbb{Z} \\ qb \in \mathbb{Z} \end{matrix}$

20

21 $\therefore x$ is rational.

22

23 By contrapositive, if x is irrational,
 24 $a+bx$ is irrational.

25

26

iii) For contradiction, assume $\sqrt{3}$ is rational.

$$\text{i.e. } \sqrt{3} = \frac{p}{q} \quad p, q \in \mathbb{Z}, q \neq 0, \text{HCF}(p, q) = 1.$$

$$\therefore 3 = \frac{p^2}{q^2}$$

$$p^2 = 3q^2$$

$\therefore p^2$ is divisible by 3

$\therefore p$ is divisible by 3 (3 is prime).

$$\text{i.e. } p = 3k, k \in \mathbb{Z}.$$

$$(3k)^2 = 3q^2$$

$$q^2 = 3k^2$$

As above, q is divisible by 3.

This contradicts $\text{HCF}(p, q) = 1$.

$\therefore \sqrt{3}$ is irrational.

By ii) $5 + 7\sqrt{3}$ is irrational.

1

$$2 \quad c) \quad x^2 - 2x + y^2 + z^2 + 4z + 4 = 0$$

3

$$4 \quad x^2 - 2x + (-1)^2 + y^2 + z^2 + 4z + (2)^2 = -4 + 1 + 4$$

5

$$6 \quad (x-1)^2 + (y-0)^2 + (z+2)^2 = 1$$

7 $\therefore$ Centre $(1, 0, -2)$. Radius of 1.

8

$$9 \quad \text{Distance of centre from origin: } \sqrt{1^2 + 2^2}$$

$$10 \quad = \sqrt{5} > 1.$$

11 $\therefore$ The origin lies outside the sphere.

12

$$13 \quad d) \quad L_1: \underline{r}_1 = \begin{bmatrix} 1 \\ -2 \\ 4 \end{bmatrix} + \lambda \begin{bmatrix} 4 \\ 5 \\ -6 \end{bmatrix}$$

14

15

$$16 \quad L_2: \underline{r}_2 = \begin{bmatrix} 3 \\ 8 \\ -2 \end{bmatrix} + \mu \begin{bmatrix} a-3 \\ -2-8 \\ 10-2 \end{bmatrix}$$

17

$$18 \quad = \begin{bmatrix} 3 \\ 8 \\ -2 \end{bmatrix} + \mu \begin{bmatrix} a-3 \\ -10 \\ 12 \end{bmatrix}$$

19

20

$$21 \quad \therefore L_1 \parallel L_2 \text{ if } \begin{bmatrix} a-3 \\ -10 \\ 12 \end{bmatrix} = \alpha \begin{bmatrix} 4 \\ 5 \\ -6 \end{bmatrix}$$

22

23

$$24 \quad \text{So } -10 = 5\alpha, \text{ then } a-3 = 4\alpha$$

$$25 \quad \alpha = -2, \quad a-3 = 4(-2)$$

$$26 \quad a = -5$$

1

2

$$e) i) \int_0^a f(x) dx \quad \text{Let } u = a - x \quad x = a - u$$

3

$$du = -dx$$

4

$$x=0 \rightarrow u=a$$

5

$$= \int_a^0 f(a-u) -du$$

$$x=a \rightarrow u=0.$$

6

7

$$= \int_0^a f(a-u) du$$

8

9

10

$$= \int_0^a f(a-x) dx$$

11

12

$$ii) \int_0^1 x(1-x)^{50} dx$$

13

14

$$= \int_0^1 (1-x) x^{50} \text{ by :)}$$

15

16

$$= \int_0^1 x^{50} - x^{51}$$

17

18

$$= \left[\frac{x^{51}}{51} - \frac{x^{52}}{52} \right]_0^1$$

19

20

$$= \frac{1}{51} - \frac{1}{52}$$

21

22

$$= \frac{1}{2652}$$

23

24

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26

15.

$$a) i) \frac{d}{dx} \left(\frac{1}{2} v^2 \right) = -\frac{1}{2} e^{-4x}$$

$$\frac{1}{2} v^2 = \frac{1}{8} e^{-4x} + c_1$$

4

$$\text{At } x=0, t=0, v=\frac{1}{2}$$

6

$$\therefore \frac{1}{2} \times \left(\frac{1}{2} \right)^2 = \frac{1}{8} e^0 + c_1$$

8

$$c_1 = 0$$

9

$$\frac{1}{2} v^2 = \frac{1}{8} e^{-4x}$$

11

$$v^2 = \frac{1}{4} e^{-4x}$$

13

$$v = \frac{1}{2} e^{-2x} \text{ as } v > 0 \text{ initially.}$$

15

$$ii) \frac{dx}{dt} = \frac{1}{2} e^{-2x}$$

17

$$\frac{dt}{dx} = 2e^{2x}$$

19

$$t = e^{2x} + c_2$$

20

$$\text{At } t=0, x=0$$

21

$$0 = e^0 + c_2$$

22

$$c_2 = -1$$

23

24

$$\therefore t = e^{2x} - 1$$

25

26

1

2

$$e^{2x} = t+1$$

3

$$2x = \ln(t+1)$$

4

$$x = \frac{1}{2} \ln(t+1)$$

5

6

iii) As $t \rightarrow 0$

7

$$x = \ln(t+1)$$

8

$$\rightarrow \infty$$

9

$$\therefore v = \frac{1}{2} e^{-2x}$$

10

$$\rightarrow \frac{1}{2} \times 0 = 0$$

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$$b) \quad i) I_n = \int_0^{\frac{\pi}{2}} x^n \cos x \, dx \quad \begin{array}{ll} u = x^n & v' = \cos x \\ u' = nx^{n-1} & v = \sin x \end{array}$$

$$= \left[x^n \sin x \right]_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} nx^{n-1} \sin x \, dx$$

$$= \left(\frac{\pi}{2}\right)^n \times \sin \frac{\pi}{2} - 0 - n \int_0^{\frac{\pi}{2}} x^{n-1} \sin x \, dx$$

$$\begin{array}{ll} u = x^{n-1} & v' = \sin x \\ u' = (n-1)x^{n-2} & v = -\cos x \end{array}$$

$$= \left(\frac{\pi}{2}\right)^n - n \left(\left[-x^{n-1} \cos x \right]_0^{\frac{\pi}{2}} + \int_0^{\frac{\pi}{2}} (n-1)x^{n-2} \cos x \, dx \right)$$

$$= \left(\frac{\pi}{2}\right)^n - n \left(0 - 0 + (n-1) I_{n-2} \right)$$

$$= \left(\frac{\pi}{2}\right)^n - n(n-1) I_{n-2}$$

$$ii) I_4 = \left(\frac{\pi}{2}\right)^4 - 4 \times 3 \times I_2$$

$$= \left(\frac{\pi}{2}\right)^4 - 12 \left(\left(\frac{\pi}{2}\right)^2 - 2 \times 1 \times I_0 \right)$$

$$= \left(\frac{\pi}{2}\right)^4 - 12 \left(\frac{\pi}{2}\right)^2 + 24 \int_0^{\frac{\pi}{2}} \cos x \, dx$$

$$= \left(\frac{\pi}{2}\right)^4 - 12 \left(\frac{\pi}{2}\right)^2 + 24 \left(\sin \frac{\pi}{2} - \sin 0 \right)$$

$$= \left(\frac{\pi}{2}\right)^4 - 12 \left(\frac{\pi}{2}\right)^2 + 24$$

$$1 \quad \text{c) } \therefore BE \perp AC$$

2

$$3 \quad \therefore \underline{\underline{b}} \perp (\underline{\underline{a}} - \underline{\underline{c}})$$

$$4 \quad \underline{\underline{b}} \perp (\underline{\underline{a}} - \underline{\underline{c}})$$

5

$$6 \quad \therefore \underline{\underline{b}} \cdot (\underline{\underline{a}} - \underline{\underline{c}}) = 0 \quad \text{--- (1)}$$

7

$$8 \quad \text{Similarly } CF \perp AB$$

$$9 \quad \underline{\underline{c}} \perp (\underline{\underline{b}} - \underline{\underline{a}})$$

$$10 \quad \underline{\underline{c}} \perp (\underline{\underline{b}} - \underline{\underline{a}})$$

11

$$12 \quad \underline{\underline{c}} \cdot (\underline{\underline{b}} - \underline{\underline{a}}) = 0 \quad \text{--- (2)}$$

13

$$14 \quad \text{ii) From (1) } \underline{\underline{b}} \cdot \underline{\underline{c}} - \underline{\underline{a}} \cdot \underline{\underline{b}} = 0$$

$$15 \quad \underline{\underline{b}} \cdot \underline{\underline{c}} = \underline{\underline{a}} \cdot \underline{\underline{b}}$$

16

$$17 \quad \text{From (2) } \underline{\underline{c}} \cdot \underline{\underline{b}} - \underline{\underline{a}} \cdot \underline{\underline{c}} = 0$$

$$18 \quad \underline{\underline{b}} \cdot \underline{\underline{c}} = \underline{\underline{a}} \cdot \underline{\underline{c}}$$

19

$$20 \quad \therefore \underline{\underline{a}} \cdot \underline{\underline{b}} = \underline{\underline{a}} \cdot \underline{\underline{c}}$$

$$21 \quad \underline{\underline{a}} \cdot \underline{\underline{b}} - \underline{\underline{a}} \cdot \underline{\underline{c}} = 0$$

$$22 \quad \underline{\underline{a}} \cdot (\underline{\underline{b}} - \underline{\underline{c}}) = 0$$

$$23 \quad \therefore \overrightarrow{AO} \perp \overrightarrow{BC}$$

$$24 \quad \overrightarrow{AO} \perp \overrightarrow{BC}$$

25

26 iii) Altitudes of a triangle are concurrent.

16.

a)

$$i) LHS = (1 + i \tan \theta)^n + (1 - i \tan \theta)^n$$

$$= \left(1 + \frac{i \sin \theta}{\cos \theta}\right)^n + \left(1 - \frac{i \sin \theta}{\cos \theta}\right)^n$$

$$= \left(\frac{\cos \theta + i \sin \theta}{\cos \theta}\right)^n + \left(\frac{\cos \theta - i \sin \theta}{\cos \theta}\right)^n$$

$$= \frac{(\cos \theta + i \sin \theta)^n + (\cos(-\theta) + i \sin(-\theta))^n}{\cos^n \theta}$$

$$= \frac{\cos n\theta + i \cancel{\sin n\theta} + \cos(-n\theta) + i \cancel{\sin(-n\theta)}}{\cos^n \theta}$$

$$= \frac{2 \cos(n\theta)}{\cos^n \theta}$$

$$ii) \text{ If } (1+z)^4 + (1-z)^4 = 0$$

$$\text{Let } z = i \tan \theta, \text{ by i):}$$

$$\frac{2 \cos 4\theta}{\cos^4 \theta} = 0$$

$$\cos 4\theta = 0$$

$$4\theta = \frac{\pi}{2}, \frac{3\pi}{2}, -\frac{\pi}{2}, -\frac{3\pi}{2} \quad \text{Need 4 solutions.}$$

$$\theta = \pm \frac{\pi}{8}, \pm \frac{3\pi}{8}$$

$$\therefore z = \pm i \tan \frac{\pi}{8}, \pm i \tan \frac{3\pi}{8}$$

b) $\therefore a=6, b=10, c=15.$

ii) Test $n=2.$

$$X_1 = \frac{2(2-1)}{2}$$

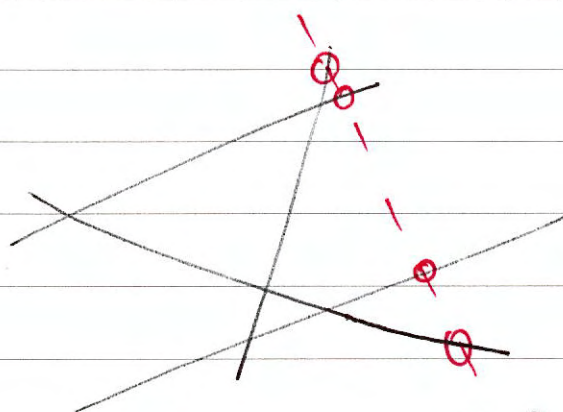
$=1$, as on the table.

$\therefore$ Statement true for $n=1.$

Assume true for $n=k$, i.e. $X_k = \frac{k(k-1)}{2}.$

Hence show true for $n=k+1$, RTP $X_{k+1} = \frac{(k+1)k}{2}$

Now, the $(k+1)^{th}$ line will maximally intersect k lines:



Intersects with every available line.

$$\begin{aligned} \therefore X_{k+1} &= X_k + k = \frac{k(k-1)}{2} + k, \text{ by assumption} \\ &= \frac{k^2 - k + 2k}{2} \\ &= \frac{(k+1)k}{2}, \end{aligned}$$

as required.

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$\therefore$ If the statement is true for $n=k$, it is true for $n=k+1$. As it is true for $n=1$, it is true for $n=1+1=2, 3, \dots$

Hence by mathematical induction, it is true for all positive integers.

1

$$2 \quad c) :) \left(\sqrt{x} - \frac{1}{\sqrt{x}} \right)^2 \geq 0$$

$$3 \quad x - 2 + \frac{1}{x} \geq 0$$

$$4 \quad x + \frac{1}{x} \geq 2.$$

5

$$6 \quad \therefore) (a+b+c) \left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \right)$$

$$7 \quad = 1 + 1 + 1 + \frac{a}{b} + \frac{b}{a} + \frac{a}{c} + \frac{c}{a} + \frac{b}{c} + \frac{c}{b}$$

$$8 \quad = 3 + \frac{a}{b} + \frac{1}{\left(\frac{a}{b}\right)} + \frac{a}{c} + \frac{1}{\left(\frac{a}{c}\right)} + \frac{b}{c} + \frac{1}{\left(\frac{b}{c}\right)}$$

10

$$11 \quad \geq 3 + 2 + 2 + 2 \quad \text{by :)}$$

$$12 \quad = 9.$$

13

$$14 \quad \text{iii) In ii), let } a = a+b, b = a+c, c = b+c:$$

$$15 \quad (a+b) + (a+c) + (b+c) \left(\frac{1}{a+b} + \frac{1}{a+c} + \frac{1}{b+c} \right) \geq 9$$

$$17 \quad 2(a+b+c) \left(\frac{1}{a+b} + \frac{1}{a+c} + \frac{1}{b+c} \right) \geq 9$$

$$18 \quad \left(\frac{a}{a+b} + \frac{b}{a+b} \right) + \left(\frac{a}{a+c} + \frac{c}{a+c} \right) + \left(\frac{b}{b+c} + \frac{c}{b+c} \right)$$

$$20 \quad + \frac{a}{b+c} + \frac{b}{a+c} + \frac{c}{a+b} \geq \frac{9}{2}$$

$$22 \quad 1 + 1 + 1 + \frac{a}{b+c} + \frac{b}{a+c} + \frac{c}{a+b} \geq \frac{9}{2}$$

$$24 \quad \frac{a}{b+c} + \frac{b}{a+c} + \frac{c}{a+b} \geq \frac{3}{2}$$

25

26